

Week 12 Reading

Multi-Objective Optimization

1 Overview

S. Yang, L. Ruangpan, A. S. Torres, and Z. Vojinovic [1]

2 Discussion Questions

1. What are the major challenges with multi-objective optimization?
2. Why do the authors model multiple NBS in each Scenario? Is this appropriate?
3. What types of tradeoffs were identified between different co-benefits?
4. Beyond the different benefits, how does the choice of optimization objective change the “best” solution (See Fig. 6)?

Before class, think of one discussion question and be ready to present.

Bibliography

- [1] S. Yang, L. Ruangpan, A. S. Torres, and Z. Vojinovic, “Multi-Objective Optimisation Framework for Assessment of Trade-Offs between Benefits and Co-benefits of Nature-based Solutions,” *Water Resources Management*, vol. 37, no. 6–7, pp. 2325–2345, May 2023, doi: 10.1007/s11269-023-03470-8.